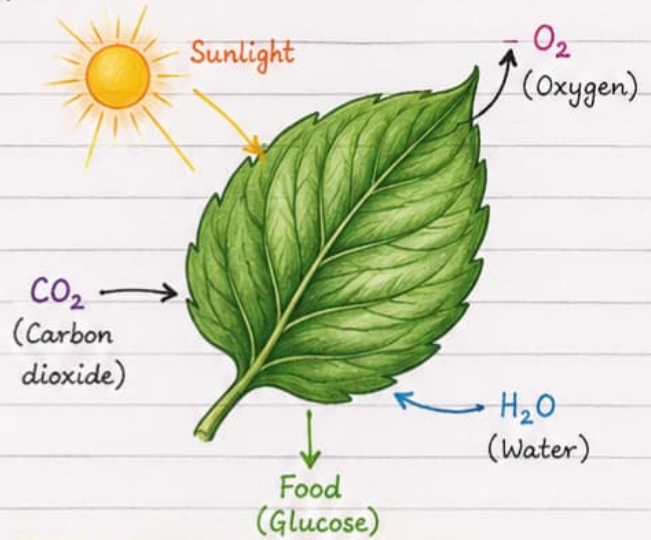


Photosynthesis

IN HIGHER PLANTS

INTRODUCTION

- Photosynthesis is the process by which green plants, algae and some bacteria prepare their own food using **light energy**.
- It occurs in **chloroplasts** and involves conversion of light energy into chemical energy.
- It is the **primary source of food** for almost all living organisms on Earth.



IMPORTANCE

- Provides food (Glucose) to plants.
- Releases **oxygen**, essential for respiration of living beings.
- Maintains balance of O_2 and CO_2 in the atmosphere.
- Forms the base of food chain.
- Helps in maintaining **ecological balance**.

PHOTOSYNTHETIC PIGMENTS

Pigments are the substances that absorb light energy for photosynthesis.

- ★ **Chlorophyll 'a'** - Primary pigment, blue-green colour
- ★ **Chlorophyll 'b'** - Accessory pigment, yellow-green colour
- ★ **Carotenoids** - Yellow, orange pigments (e.g., Carotene, Xanthophyll)
- ★ **Xanthophyll** - Helps in protecting chlorophyll from photo-oxidation.

SITE OF PHOTOSYNTHESIS

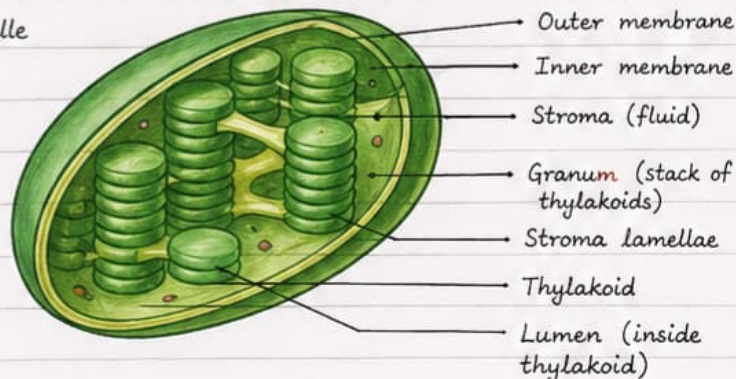
Main site - **Chloroplast**

Chloroplast - A double membraned organelle that contains chlorophyll and other pigments.

PARTS OF CHLOROPLAST

- Outer membrane** - Smooth, permeable
- Inner membrane** - Regulates entry of substances
- Stroma** - Fluid matrix containing enzymes, DNA, ribosomes
- Thylakoids** - Flattened sacs containing chlorophyll
- Grana** - Stacks of thylakoids
- Stroma lamellae** - Unstacked thylakoids connecting grana

STRUCTURE OF CHLOROPLAST



Chlorophyll is present in the **thylakoid membrane**, while enzymes of Calvin cycle are present in the **stroma**.

PROCESS OF PHOTOSYNTHESIS

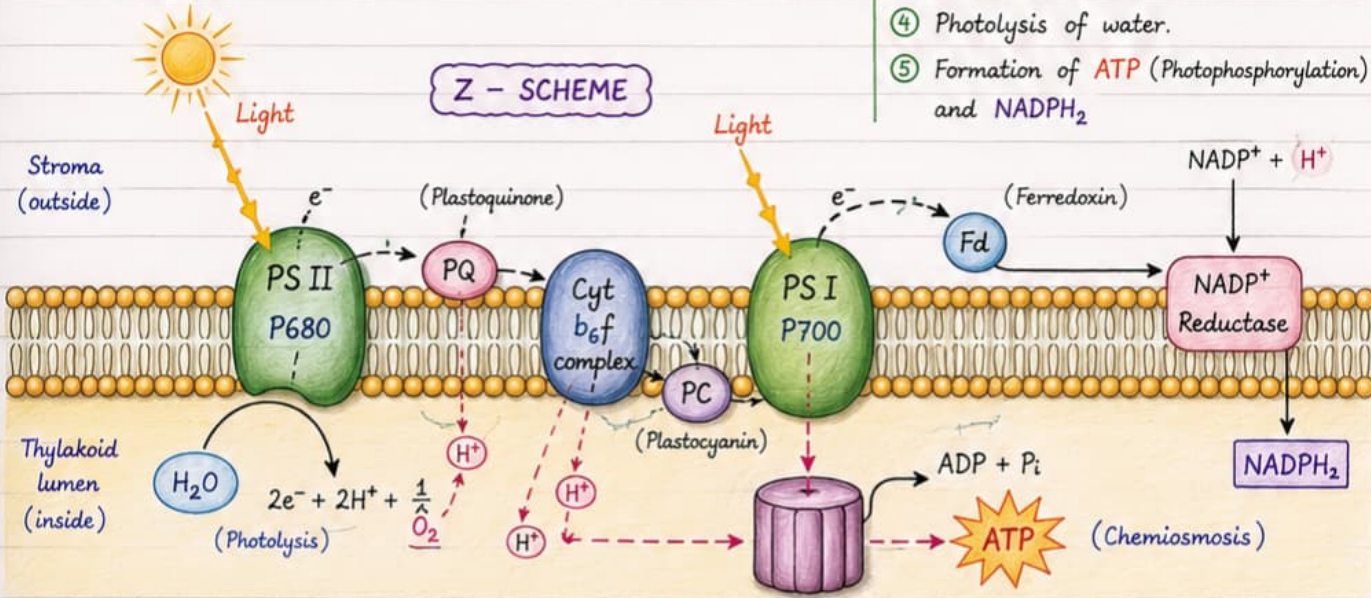
Photosynthesis occurs in two major stages - ① Light dependent reactions - ② Light independent reactions (Dark reactions / Calvin cycle).

1 LIGHT DEPENDENT REACTIONS (Occurs in Thylakoid Membrane)

KEY STEPS

- Requires light energy.
- Water is split (Photolysis) releasing O_2 , H^+ and e^- .
- Produces **ATP** and **NADPH₂** which are used in dark reactions.

- ① Absorption of light by chlorophyll.
- ② Excitation of electrons.
- ③ Flow of electrons through electron transport chain.
- ④ Photolysis of water.
- ⑤ Formation of **ATP** (Photophosphorylation) and **NADPH₂**

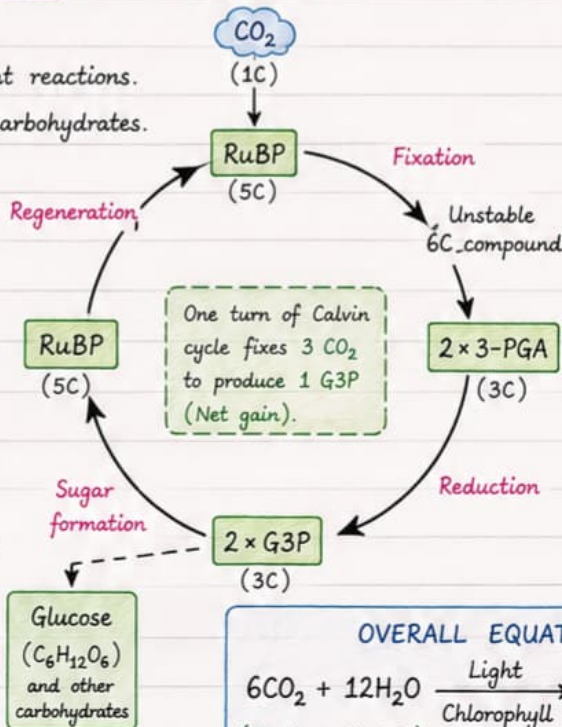


2 DARK REACTIONS (CALVIN CYCLE) (Occurs in Stroma)

- Does not require light directly.
- Uses **ATP** and **NADPH₂** from light reactions.
- CO₂ is fixed and converted into carbohydrates.

STEPS OF CALVIN CYCLE

- ① **CARBON FIXATION** - CO₂ combines with RuBP (Ribulose-1,5-bisphosphate) with the help of enzyme RuBisCO forming an unstable 6C compound which breaks into two molecules of 3-PGA.
- ② **REDUCTION** - 3-PGA is reduced to G3P (Glyceraldehyde-3-phosphate) using ATP and NADPH₂.
- ③ **REGENERATION** - Some G3P is used to regenerate RuBP using ATP.
- ④ **SUGAR FORMATION** - Remaining G3P molecules combine to form glucose and other carbohydrates.



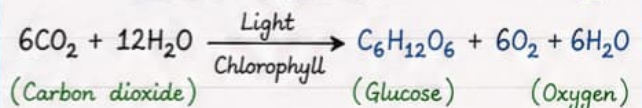
INPUTS

- CO₂
- ATP
- NADPH₂

OUTPUTS

- G3P
- ADP + P_i
- NADP⁺

OVERALL EQUATION



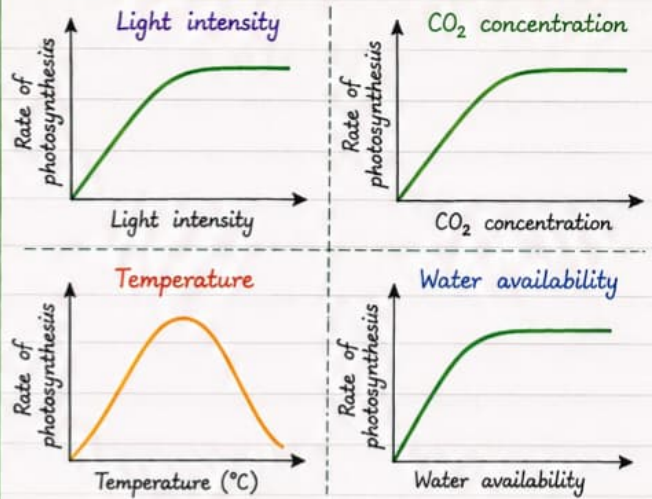
Calvin cycle is also called **C₃ cycle** because the first stable product is a **3-carbon** compound.

★ **RUBISCO** - The most abundant enzyme on Earth, plays a key role in carbon fixation.

FACTORS AFFECTING PHOTOSYNTHESIS

- LIGHT** - Rate increases with increase in light intensity up to a saturation point. Beyond this, rate remains constant.
- CO₂ CONCENTRATION** - Increase in CO₂ concentration increases rate up to a point, then becomes constant.
- TEMPERATURE** - Optimum temperature for most plants is 25°-35°C. Very low or high temperature decreases the rate.
- WATER** - Water is essential as it is the raw material. Deficiency causes stomatal closure and decreases rate.
- CHLOROPHYLL CONTENT** - More chlorophyll, more is the photosynthetic rate.

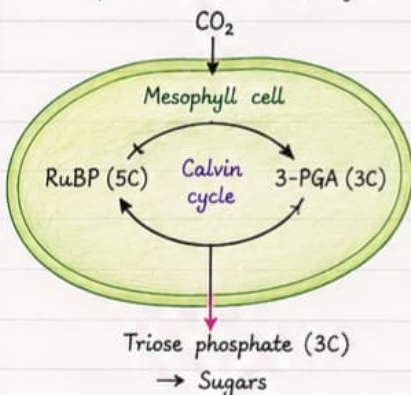
RATE vs FACTORS (Graphical Representation)



CO₂ FIXATION PATHWAYS

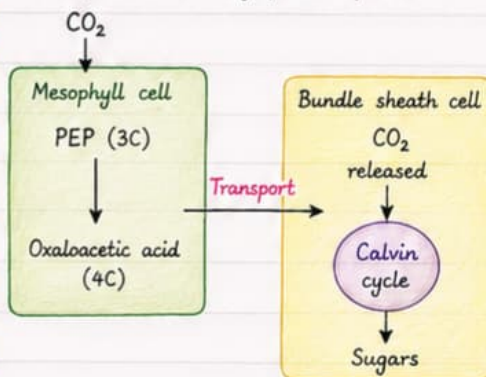
C₃ PATHWAY (Calvin cycle only)

- ★ Occurs in all plants.
- ★ First stable product is 3-PGA (3C).
- ★ Example - Rice, Wheat, Soybean.



C₄ PATHWAY (Hatch & Slack pathway)

- ★ Occurs in some plants (maize, sugarcane, grasses).
- ★ First stable product is Oxaloacetic acid (4C).
- ★ Helps in reducing photorespiration in hot climate.

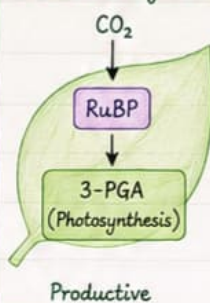


- ADVANTAGE OF C₄ PLANTS**
- Minimizes photorespiration.
 - More efficient in hot, dry climates.
 - Higher productivity.

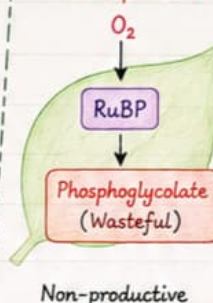
PHOTORESPIRATION

- It is a wasteful process in which RuBP reacts with O₂ instead of CO₂ in the presence of RuBisCO.
- It leads to the formation of phosphoglycolate which consumes energy and releases CO₂.
- Occurs more in C₃ plants, high temperature, and low CO₂ concentration.

Normal Photosynthesis



Photorespiration



SIGNIFICANCE

- ✓ Provides food and energy for all living organisms.
- ✓ Maintains O₂ and CO₂ balance in the atmosphere.
- ✓ Basis of all food chains.
- ✓ Essential for survival of life on Earth.
- ✓ Helps in maintaining ecological balance.

IMPORTANT POINTS

- Chlorophyll captures light but does not use energy.
- ATP and NADPH₂ are energy carriers.
- Calvin cycle does not require light directly but depends on ATP and NADPH₂.

DID YOU KNOW?

If photosynthesis stops for just a few days, life on Earth will collapse due to lack of food and oxygen.



MEMORY TRICK

Light Reaction → ATP + NADPH₂
Dark Reaction → Sugar

"LIGHT makes ENERGY,
DARK makes FOOD."

MECHANISM OF PHOTOSYNTHESIS

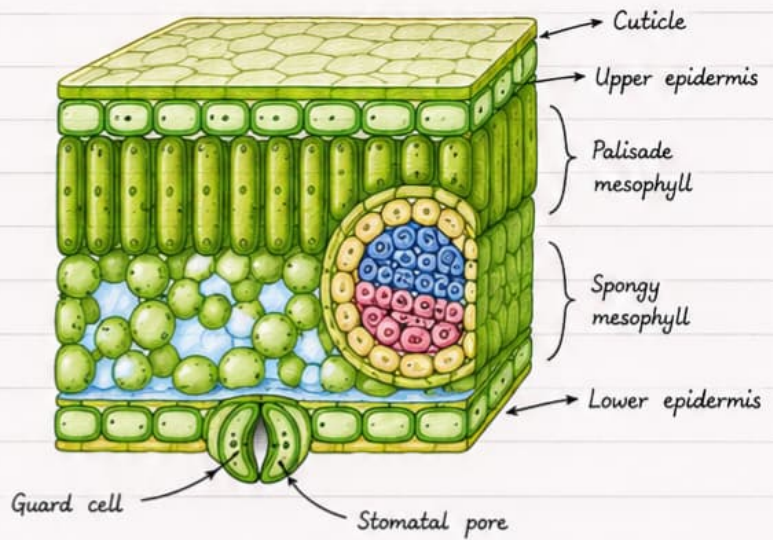
OVERVIEW

- Photosynthesis is a complex process involving light absorption, conversion of light energy to chemical energy and CO_2 fixation.
- It occurs in **chloroplasts** of green cells.
- It has two major phases -
Light reactions (in **thylakoid membrane**) and **Dark reactions** (in **stroma**).

ROLE OF STOMATA

- Allows CO_2 to enter the leaf.
- Allows O_2 and water vapour to exit.
- Regulates gaseous exchange and transpiration.

INTERNAL STRUCTURE OF A LEAF (T.S.)



EXPERIMENTAL EVIDENCES FOR PHOTOSYNTHESIS

1 STARCH TEST

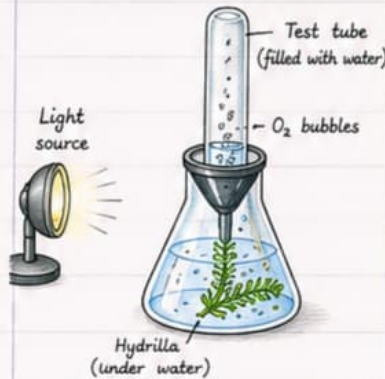
A leaf exposed to sunlight forms starch.



Blue-black colour appears in the part exposed to light → Starch is formed.

2 HILL'S EXPERIMENT

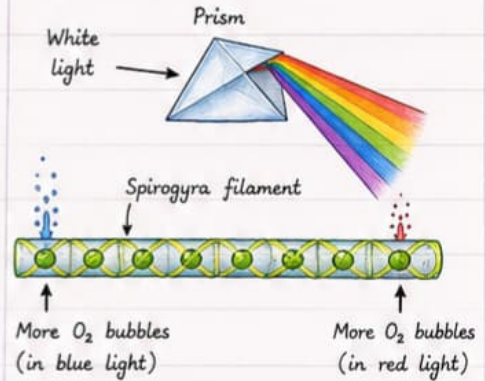
Light is necessary for the formation of O_2 .



O_2 bubbles are released only in the presence of light.

3 ENGELMANN'S EXPERIMENT

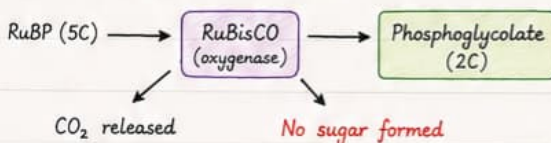
Photosynthesis is maximum in red and blue light.



Maximum photosynthesis occurs in blue and red regions of spectrum.

PHOTORESPIRATION

- Occurs in presence of light but in the absence of CO_2 .
- RuBisCO acts as an oxygenase.
- O_2 is taken in and CO_2 is released.
- Consumes O_2 and reduces efficiency of photosynthesis.
- Common in C_3 plants and in high temperature.



SUMMARY

- 1 Photosynthesis takes place in **chloroplasts** of green plants.
- 2 It uses light energy to convert CO_2 and H_2O into glucose and O_2 .
- 3 Occurs in two stages - Light reactions (in **thylakoid membrane**) and Dark reactions (Calvin cycle in **stroma**).
- 4 Light, CO_2 , temperature, water and chlorophyll are key factors affecting photosynthesis.
- 5 It is the basis of life on Earth and sustains food chains.



Remember: Without photosynthesis, there would be no food, no **oxygen** and no life on Earth.

LIGHT is the **SOURCE**, **PLANTS** are the **PRODUCERS**, **LIFE** is the **RESULT**.

